

cmvt: A Python toolkit for reproducible external validation of clinical prediction models

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Software

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Summary

Clinical prediction models are routinely reported with a single external-validation headline: AUROC fell from some internal value to some lower external value. cmvt (clinical-model-validation-toolkit) is a Python package that treats that number as insufficient on its own and provides the machinery to unpack it: DeLong-based comparison of correlated AUROCs (DeLong et al., 1988), calibration diagnostics and recalibration, patient-level cluster bootstrap confidence intervals, distribution-shift metrics (population stability index, KL divergence, energy distance, and RBF-kernel MMD), decidable data-integrity checks (patient overlap across splits, duplicate identifiers, post-cutoff leakage, exact label duplication, and Little's MCAR test), and a validation engine that decomposes an external-validation AUROC gap into a component attributable to true distribution shift and a component attributable to information loss from features the external site never recorded.

The package ships a synthetic three-cohort generator (internal, temporal, external) so the full pipeline runs end-to-end with no external data dependency and no network access: `pip install -e ".[dev]"` followed by `cmvt-demo` builds roughly 12,000 synthetic patients, runs every module, and writes a Markdown validation report in under a minute on a laptop. All 92 tests in the test suite assert numerical agreement against an independent reference implementation where one exists (scipy, statsmodels, scikit-learn) rather than only checking that the code runs, and continuous integration exercises the suite on Python 3.10, 3.11, and 3.12 with a 90% minimum-coverage gate.

Statement of need

External validation is a required step before a clinical prediction model is trusted outside the population it was developed on (Debray et al., 2015; Steyerberg, 2019), and calibration failures in particular are common enough to be called predictive analytics' "Achilles heel" (Van Calster et al., 2019). Two distinct explanations are typically available whenever external AUROC is lower than internal AUROC: the external population genuinely differs from the development population (distribution shift) (Finlayson et al., 2021; Subbaswamy & Saria, 2020), or the external site did not record some of the predictors the model was trained on, so those predictors are zero-imputed or dropped at deployment, discarding real signal. These two explanations call for different remedies — model updating or population-specific redevelopment for genuine shift, versus better data capture or a model redesigned around what is actually available for information loss — but a single external AUROC number cannot distinguish them.

`cmvt.engine.validation_plan` addresses this directly: it fits the development model on its full recorded feature set and evaluates it externally with the external site's missing features zero-imputed (the "naive" transfer), separately retrains on only the features every registered cohort actually shares and evaluates that ("fair") model externally, and reports the difference between the naive and fair AUROC gaps as an information-loss confound (Figure 1). Existing general-

purpose model-validation and fairness toolkits provide calibration and subgroup-performance diagnostics but do not, to our knowledge, provide this specific decomposition, nor construct synthetic cohorts with a known ground-truth split between shift and missingness against which such a decomposition could be checked for correctness during development.

A second design goal shapes the package as much as its calculations: `cmvt` is built to show what it will not compute. Three modules ship alongside executable negative controls (`tests/test_negative_controls.py`) demonstrating, on data constructed so the full-data truth is known, that (1) MAR-versus-MNAR status is not identifiable from observed data alone, so `cmvt.integrity.littles_mcar_test` tests only for MCAR and does not extend to an MNAR-detection claim; (2) a composite “dataset quality” score built from completeness, balance, size, and shift-freedom crowns a different dataset as “best” depending on the (equally defensible) weighting chosen, so `cmvt` reports sub-metrics individually and does not rank them; and (3) a values-only leakage detector cannot distinguish a legitimate predictor from an outcome-derived leak with identical AUROC, correlation, and mean difference against the label, so `cmvt.integrity.leakage_scan` restricts itself to checks that are decidable from the data alone (split overlap, duplicate IDs, post-cutoff rows, exact-label-duplicate columns). This is intended for researchers and clinical-informatics practitioners who need not just a working validation pipeline but one whose claims are explicitly bounded by what is statistically identifiable.

Figure 3. Validation engine — separating true distribution shift from information loss

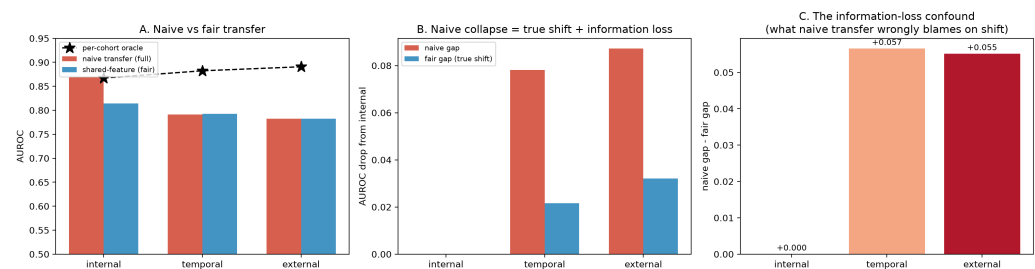


Figure 1: Output of the bundled synthetic demo (`cmvt-demo`), reproduced deterministically by `scripts/generate_figures.py`. Panel A compares naive full-feature transfer against shared-feature (“fair”) transfer across three synthetic cohorts of increasing dissimilarity, against a per-cohort oracle ceiling. Panel B shows the naive AUROC drop from the internal cohort decomposing into a true-shift component (blue, the fair gap) and an information-loss component (the remainder). Panel C isolates that information-loss confound directly.

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References

- Debray, T. P. A., Vergouwe, Y., Koffijberg, H., Nieboer, D., Steyerberg, E. W., & Moons, K. G. M. (2015). A new framework to enhance the interpretation of external validation studies of clinical prediction models. *Journal of Clinical Epidemiology*.
- DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*.
- Finlayson, S. G., Subbaswamy, A., Singh, K., & others. (2021). The clinician and dataset shift in artificial intelligence. *New England Journal of Medicine*.

- 73 Steyerberg, E. W. (2019). *Clinical prediction models: A practical approach to development,*
74 *validation, and updating* (2nd ed.). Springer.
- 75 Subbaswamy, A., & Saria, S. (2020). From development to deployment: Dataset shift, causality,
76 and shift-stable models in health AI. *Biostatistics*.
- 77 Van Calster, B., McLernon, D. J., Smeden, M. van, Wynants, L., & Steyerberg, E. W. (2019).
78 Calibration: The achilles heel of predictive analytics. *BMC Medicine*.

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